



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE– (24AD3107)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	"HealthScope" – A Full-Lifecycle Public Health Analytics & Risk Dashboard	
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Abstract:

Chronic diseases such as diabetes and heart disease are major public health concerns that can be influenced by various factors including age, blood pressure, glucose levels, BMI, cholesterol, and lifestyle indicators. Identifying the relationships between these health factors and disease outcomes is important for understanding population health patterns and supporting early risk identification. However, healthcare datasets often contain large amounts of information that can be difficult to analyze manually. Traditional analysis methods may also make it difficult to identify complex patterns and relationships between multiple health-related factors. This highlights the need for a data-driven system that can efficiently analyze healthcare data and provide meaningful insights into disease risk.

To address this need, HealthScope – A Full-Lifecycle Public Health Analytics & Risk Dashboard is proposed as a public health analytics and risk prediction platform that applies data science and machine learning techniques to healthcare datasets. The system uses publicly available datasets related to diseases such as diabetes or heart disease. The collected data is first subjected to data preprocessing to improve its quality and suitability for analysis. Exploratory Data Analysis (EDA), statistical analysis,

and feature engineering are then performed to identify important patterns, relationships, and potential risk factors within the dataset.

Machine learning techniques are incorporated to develop models capable of predicting disease risk. Multiple classification algorithms, including Logistic Regression, Decision Tree, and Random Forest, are trained using the processed healthcare data. The models are evaluated using appropriate performance metrics to compare their effectiveness and identify a suitable model for risk prediction. This approach allows the system to not only analyze existing health patterns but also use machine learning to estimate the risk of disease for individual cases.

The results of the analysis and machine learning models are integrated into an interactive dashboard. The dashboard provides visual representations of health trends, analysis of major risk factors, machine learning model performance, and individual risk predictions. By combining data preprocessing, analysis, visualization, and machine learning into a single platform, HealthScope provides a complete data science workflow for public health analytics.

Collectively, HealthScope demonstrates how data science and machine learning can be applied to healthcare datasets to transform raw data into meaningful and accessible information. The proposed platform aims to support early risk identification, better understanding of public health patterns, and data-driven decision-making through an interactive and user-friendly analytics environment.

Project Architecture/Project Overview:

